LECTURE

AI in Engineering Design and CAD/CAM

06

Name the kind of AI before trusting it with a design decision.

What kind of AI designed this bracket?



An ‘organic’ openwork shape — looks ‘grown’ by a neural network

Answer:

No generative AI at all.

A deterministic numerical optimizer.

The statement — 1904.

8 parts → 1 · mass −~40%
>150 variants evaluated

GM seat bracket (General Motors × Autodesk, 2018) · the statement goes back to Michell's work, 1904.

The lecture's spine — three questions to yourself and six parts.

Three anchor questions — repeated in every part

1. **Deterministic math or a trained probabilistic model?**
2. **What is being optimized, and who set the constraints?**
3. **Who bears responsibility?**

0	1	2	3	4	5	6
Start + map	Foundation + taxonomy	Generative design ≠ genAI	Surrogate / PINN	Generative AI / LLM	When to say ‘no’	Synthesis

Classification — for all 6 classes of AI; depth — for the three with the highest cost of error.

PART

1

Name the kind of AI

the foundation — solving an equation vs. guessing from examples — and the taxonomy of six classes

Deterministic optimization solves an equation set by the engineer; a trained model guesses from examples.

SOLVES a given equation

- The engineer formulates the objective and constraints (equilibrium, strength, volume)
- The algorithm numerically searches for a point satisfying the equations
- The physical law is present explicitly
- Same input → same answer · reproducible and traceable

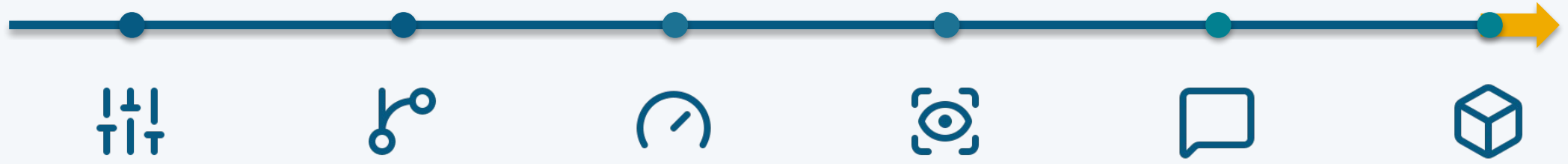
GUESSES from examples

- The model is trained on a corpus of examples
- It reproduces ‘what an answer usually looks or sounds like’
- There is no equilibrium equation inside
- An error does not self-diagnose

Any tool labeled ‘with AI’ is first placed on this axis.

Six classes of AI fall on one axis: the farther right, the fewer the built-in guarantees.

This is a reference skeleton — the six classes don't need memorizing yet: they'll fill with meaning as the lecture goes.



1. Optim. ML / topology optimization

moves material density, min. mass at σ ≤ [σ]

2. Evolutionary / GA

mutation and selection against a given objective

3. Surrogate / PINN

a network learned input→ output from thousands of runs

4. Computer Vision

a probabilistic image classifier

5. LLM assistant

text drafts, explains clauses of a standard

6. Generative AI for geometry

generates 3D with no physical guarantees

classes 1–2 — optimizing a given function · 3 — approximating a solver · 4–6 — probabilistic models

The full attribute matrix — at the end of the lecture, once every cell is filled with meaning.

Optimization ML and surrogate/PINN — recognized by the boundary.

	When YES	When NO	Classical alternative
Optimization ML / topology optimization	reducing mass at a given strength; parts for additive manufacturing	constraints set incompletely — the optimum fits the wrong statement	parametric optimization + a code-based calculation
Surrogate / PINN	fast evaluation of many variants inside the training domain	extrapolation beyond training, field discontinuities, certification	full FEA / CFD with a mesh-convergence check

Generative AI/LLM, computer vision, and evolutionary/GA — one example and boundary each.

Generative AI / LLM assistant

✓ when YES

a concept sketch, mood board; a text draft subject to review

✗ when NO

a part going into calculation/production with no rework; a source of truth on grades

↳ alternative

parametrics + det. topology optimization + FE check; norms

Computer Vision

✓ when YES

high-volume routine sorting as a first filter

✗ when NO

the sole arbiter of fitness in safety-critical NDT

↳ alternative

normative NDT with a technician and a POD methodology (probability of detection)

Evolutionary / GA

✓ when YES

conflicting requirements (NASA's ST5 antenna)

✗ when NO

certification explainability and interpretability are needed

↳ alternative

analytical / gradient-based synthesis per a proven methodology

Anchors: deterministic/probabilistic? · what's optimized, who set the constraints? · who is responsible?

PART

2

Generative design is not generative AI

> a deterministic optimizer on classical math (Michell, Cauchy, SIMP), not a generative model

‘AI generates the design’ — a marketing substitution of concepts.



Marketing says

- “AI generates the design”
- “AI came up with the part”
- “creative artificial intelligence”



What's actually happening

- Topology optimization — a deterministic numerical method
- Same input → same answer
- Not a single trained generative model
- The vendor itself distinguishes topology optimization from ‘generative AI’

Lesson: before trusting a tool — ask what kind of AI this is.

Topology optimization is gradient descent on the density of material.

This is a lineage (logical order), not a chronology: Michell's statement (1904) is the foundation; Cauchy's algorithm (1847) is older by date but is applied to it later.

logical lineage →

Known from strength of materials
underloaded material can be removed (fully-stressed design)

Michell 1904
the intuition and statement (minimum material)

Cauchy 1847 / gradient descent
the same algorithm of steepest descent

Bendsoe–Kikuchi 1988
the numerical method: continuous density

SIMP 1989
the industry standard: $\rho \in [0,1]$, a penalty toward 'black-and-white'

at every step, the equilibrium equation is explicitly solved — convergence to the optimum, NOT a sample from a distribution

KKT / Lagrange multipliers — the formal optimality condition; no formulas on this slide.

GM and Airbus: a radical mass reduction — numerical optimization and enumeration, not diffusion.

GM seat bracket

- 8 parts → 1
- mass −~40%
- >150 variants evaluated

kind of AI: numerical optimization + cloud-based enumeration

Airbus bionic partition

- mass −~45%
- ≈35 kg vs. ≈65 kg
- material Scalmalloy

kind of AI: topology/lattice optimization + enumeration

Both parts require additive manufacturing — this is a constraint, not a coincidence.

In Russia, APM FEM in KOMPAS-3D is deterministic SIMP, not AI.



Real achievements

- OKB Sukhoi — a load-bearing bracket for the Su-57, about a quarter lighter
- a digital model on a supercomputer · LPBF (laser powder bed fusion)
- a prototype was worked on for the MS-21-300



Substitution of concepts

- APM FEM excludes a substantial share of volume while preserving strength
- but it's deterministic SIMP, NOT AI — the same trap
- the C3D kernel is a strategic asset (only a handful of kernels exist worldwide)



Where we honestly lag

- no mature AI-generative engine at the level of Fusion exists in Russia
- the honest alternative — classical topology optimization
- + manual reconstruction into an editable solid

The exact share of volume excluded by APM FEM — per the vendor's claim.

The main risk of topology optimization is not a ‘hallucination’ but garbage-in → optimal garbage.

1. Incomplete statement

a load case forgotten;
manufacturing constraints not set

2. Deterministic optimizer

honestly solves the problem given

3. A flawless optimum — an unusable shape

mathematically correct, cannot be manufactured

Non-manufacturability: closed cavities and curved ribs with no cutter access — additive only; hard to measure and inspect against ESKD.

The alternative for serial milling/casting — classical parametrics + manual topology optimization for manufacturability.

PART

3

AI speeds up the math, not the physics

> a surrogate guesses the solver's output from examples; a PINN smooths out sharp jumps

FE analysis solves the equilibrium equations; a surrogate guesses the field from examples.

FE solver

- part → mesh → solves a system of equilibrium equations → stress field
- proven, traceable, deterministic math
- error controlled by a mesh-convergence check

Surrogate

- learned the mapping ‘geometry+loads → stresses’ on thousands of computations
- does NOT solve the equation — it guesses from examples
- trained offline, in advance, on an expensive archive

Under the hood is a system of equilibrium equations; the surrogate does not solve it.

Surrogates-2026: ‘up to N×’ is an upper bound, not typical.

By 2026, major vendors turned surrogates into a product — read the numbers critically.

Ansys SimAI

2026 R1 release

a surrogate on a reduced-order model; a speedup of tens to hundreds of times

Altair PhysicsAI

trained on an archive

trained on a historical simulation archive; a speedup of up to on the order of a thousand times

NVIDIA PhysicsNeMo

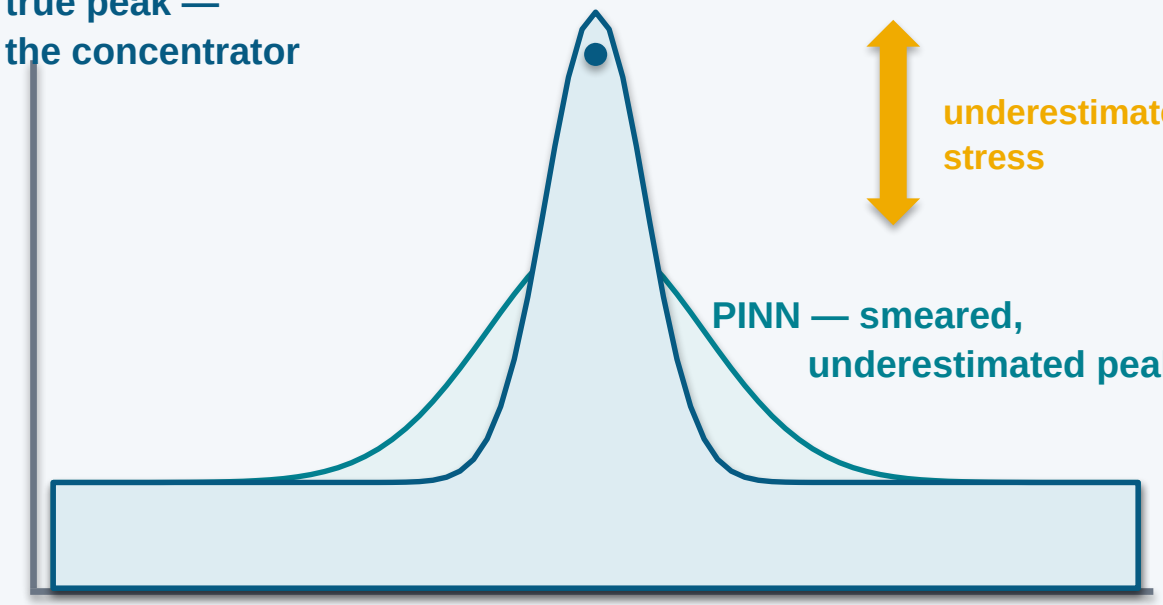
open-source

an open Python physics-AI framework: from PINNs to neural operators

‘up to ~10^2–10^3×’ (≈ ×100–×1000) — on a case favorable to the vendor. The cost of generating the training archive (thousands of expensive solver runs) is excluded from the ROI.

A PINN ‘smears’ the peak at a concentrator and gives an underestimated stress.

stress along a section near a hole / groove



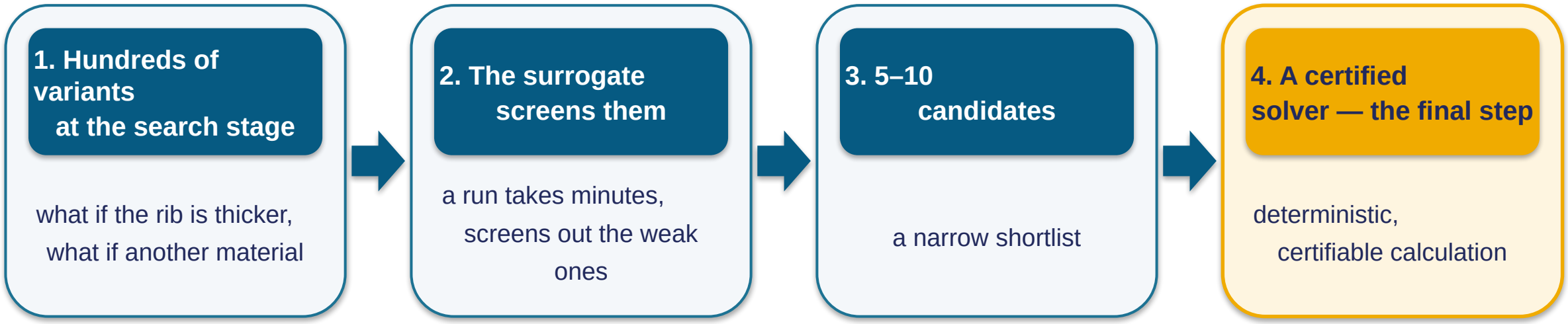
an underestimated stress in the dangerous cross-section = an underestimated risk of fracture

Other limitations

- does not outperform mature solvers on forward problems, often slower
- does not generalize beyond training
- opaque on failure

Where a PINN is genuinely useful: sparse data, inverse problems, complex geometry — NOT a replacement for FEM/CFD.

A surrogate is for enumeration; the final result is on a certified solver.



Criterion: a surrogate is valid only inside its training domain. Extrapolation = a silent error (the model does not report it has left the domain).

Russia: 'Logos' (Rosatom) — surrogates in the wrapper, not inside the solver's kernel · CML-Bench (SPbPU) — an accumulated archive → an assistant.

PART

4

Where a hallucination costs dearly

> generative AI and LLMs contain no physical law — the cost of a confident error in design is high

Mature — LLM as an interface; generating geometry from text — an early beta.

Mature layer: LLM as an interface

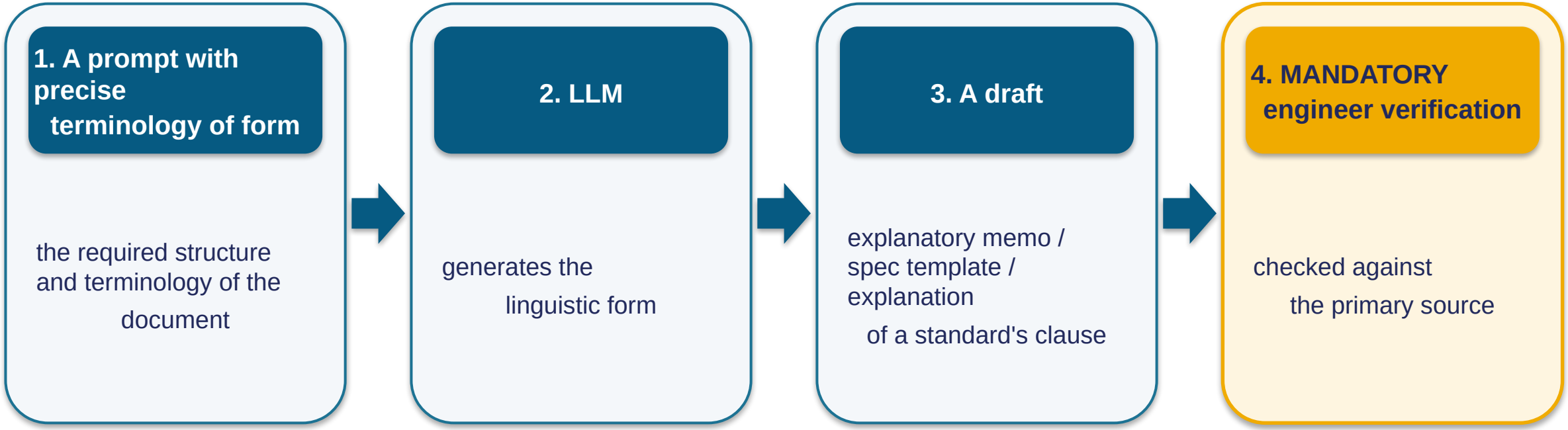
- a text prompt → CAD commands over existing deterministic operations
- under the engineer's control
- e.g. Autodesk Assistant, Siemens Design Copilot

Immature layer: text-to-CAD / neuro-CAD

- 3D geometry generation from text
- e.g. Zoo.dev, Autodesk Bernini
- commercial availability of neuro-CAD stated as ‘upcoming,’ no date

The vendor itself labels Bernini ‘strictly experimental, not for public use’ — an honest maturity signal.

An LLM writes the draft of technical text — the engineer verifies the facts.



An LLM gives the right FORM and potentially the wrong CONTENT — useful for a draft, dangerous for the final document. The boundary runs between form (the model) and content (the engineer); verification — always, without exception.

Mars Climate Orbiter: $327M — an unchecked unit mismatch at a seam between systems.



Mars Climate Orbiter (1999)

- contact lost on Mars orbit insertion · mission cost **≈ $327M**
- the ground module output pound-force-seconds, the program expected newton-seconds — a mismatch of ×4.45
- the conversion was performed by no one: each side assumed the other had done it

An LLM assistant = exactly such a seam between systems: it takes quantities from one context, passes them into another, and CONFIDENTLY confuses units (psi and MPa, mm and inches).



Gimli Glider (1983) — a manual fuel recalculation with the wrong coefficient



Hyatt Regency (1981) — a change to a suspension joint ‘by word of mouth,’ with no independent recalculation

An LLM errs on roughly a third to a half of engineering calculations.

Reproducible statistics — an argument of a different type than individual historical catastrophes.

ORCA Benchmark — 5 SOTA models on real calculations



best ~63%

accuracy ~45–63% (a range, not an exact number)

EngiBench

a drop in accuracy under a simple rephrasing = pattern-matching, not reasoning

Aerospace case

a treatment recommendation violated 3 independent standards at once — it sounded expert

An LLM is not a source of truth on grades/GOST/tolerances · independent verification at every interface with AI · a change to the force scheme = a formal change-request, not an ‘OK’ in a chat.

PART

5

When to tell AI ‘no’

> criteria of non-applicability, the right tool, and who bears responsibility

The higher the cost of error — the less room for probabilistic AI.

Criterion	Why probabilistic AI doesn't fit	The more correct tool
Certification calculation	traceability is needed	**a certified solver + verification**
Safety factor	set by the code, not the model	**a norm + engineering justification**
Code inspection GOST/ISO	hallucinates numbers and tolerances	**the normative base + an inspector**
Legal responsibility	AI is not a subject of responsibility	**an engineer's signature**
An exact solver exists	an approximation is worse than exact	**a direct numerical calculation**
Safety of life	the cost of error is lives	**a norm calc + check + NDT**
Units at a system seam	confidently confuses units (Mars)	**a dimensional check + a single SI**
A change to the force scheme	‘in a chat,’ no recalculation (Hyatt)	**a change-request + recalculation**

AI is a tool for expanding variety and drafting, not an arbiter of truth and not a bearer of responsibility.

In Russia, the boundaries of AI geometry are concrete and normative.

The ESKD barrier

- a ‘grown’ organic geometry fits poorly onto a 2D drawing
- issuing inspection-compliant documentation is non-trivial

Attestation and verification

- the APM Structure3D kernel holds a Rostekhnadzor certificate
- without certified verification, a structure isn't admitted to supervised production

Federal Law 187 / 58

- for significant CII facilities, cloud-based Western CAD doesn't apply
- the 2022–2024 AutoCAD/Fusion licensing lockout — a single point of failure

Authorship and law

- under Russian law a machine is not a subject of copyright
- the engineer or the organization is responsible, not the tool

CII — critical information infrastructure (Federal Law 187).

AI generates the shape — the human is responsible, verifies, and invents the function.

AI

- proposes a shape for a fixed function
- distributes material within a given volume

Human

- sets the constraints
- IS RESPONSIBLE for the result (a signature = legal responsibility for safety)
- the constraint is deontological, not technical — not delegable

TRIZ (Altshuller, from 1946) formalizes INVENTION — resolving a technical contradiction; this is a level above optimizing shape for a fixed function.

AI generates — the human is responsible, verifies, and invents the function.

PART

6

Synthesis

> the classes and failures are dissected — now we assemble a decision rule on a concrete problem

Lighten the bracket: one anchor question split the justified answers.

Problem: reduce the mass of a load-bearing bracket

(a) serial milling

- Q1 a guarantee is needed → deterministic
- Q2 the critical constraint is milling manufacturability; pure topology optimization yields the unmanufacturable
- Q3 certified verification + a signature are needed

Parametrics + manual topology optimization for manufacturability; the final step — a certified FEA. NOT generative AI, NOT a surrogate for the final step.

(b) additive

- Q1 the same need for a guarantee → deterministic
- Q2 additive removes shape constraints — topology optimization at full strength
- Q3 the final step is on a certified solver; a surrogate is for enumeration

Deterministic topology optimization + a final step on the solver; a surrogate is optional for enumeration; generative AI for geometry — still no.

One anchor question (Q2 — what is optimized and what the constraints are) split the answers.

The decision rule — five questions; the full matrix assembled from all the parts.

The decision rule

1. **Name the kind of AI, place it on the axis**
2. **Deterministic or probabilistic?**
3. **What's optimized, who set the constraints? Is the statement complete?**
4. **Is there an exact / normative tool?**
5. **Who is responsible?**

Full taxonomic matrix of 6 classes

Class	Nature	Guarantees	Maturity	Who's responsible
Topology optimization	determ.	optimum of a function	mature	**engineer**
Evolutionary / GA	stochastic	optimum of a function	niche	**engineer**
Surrogate / PINN	probabilistic	only inside the domain	growing	**engineer**
Computer Vision	probabilistic	a first filter	mature	**engineer**
LLM assistant	probabilistic	linguistic form	mature	**engineer**
Generative geometry	probabilistic	nothing about physics	beta	**engineer**

the 'who's responsible' column — everywhere, the engineer

Specialized departments and supercomputing centers at technical universities work across this whole spectrum.



Q&A?

If there are no questions — three provocations:

1. *Which tool does marketing call ‘AI’ in a familiar CAD system — and what kind of AI is it actually?*
2. *Where in a project is there a ‘seam between systems’ at which AI could confidently mix up units?*
3. *Name a problem where AI is justifiably told ‘no,’ and the right tool.*